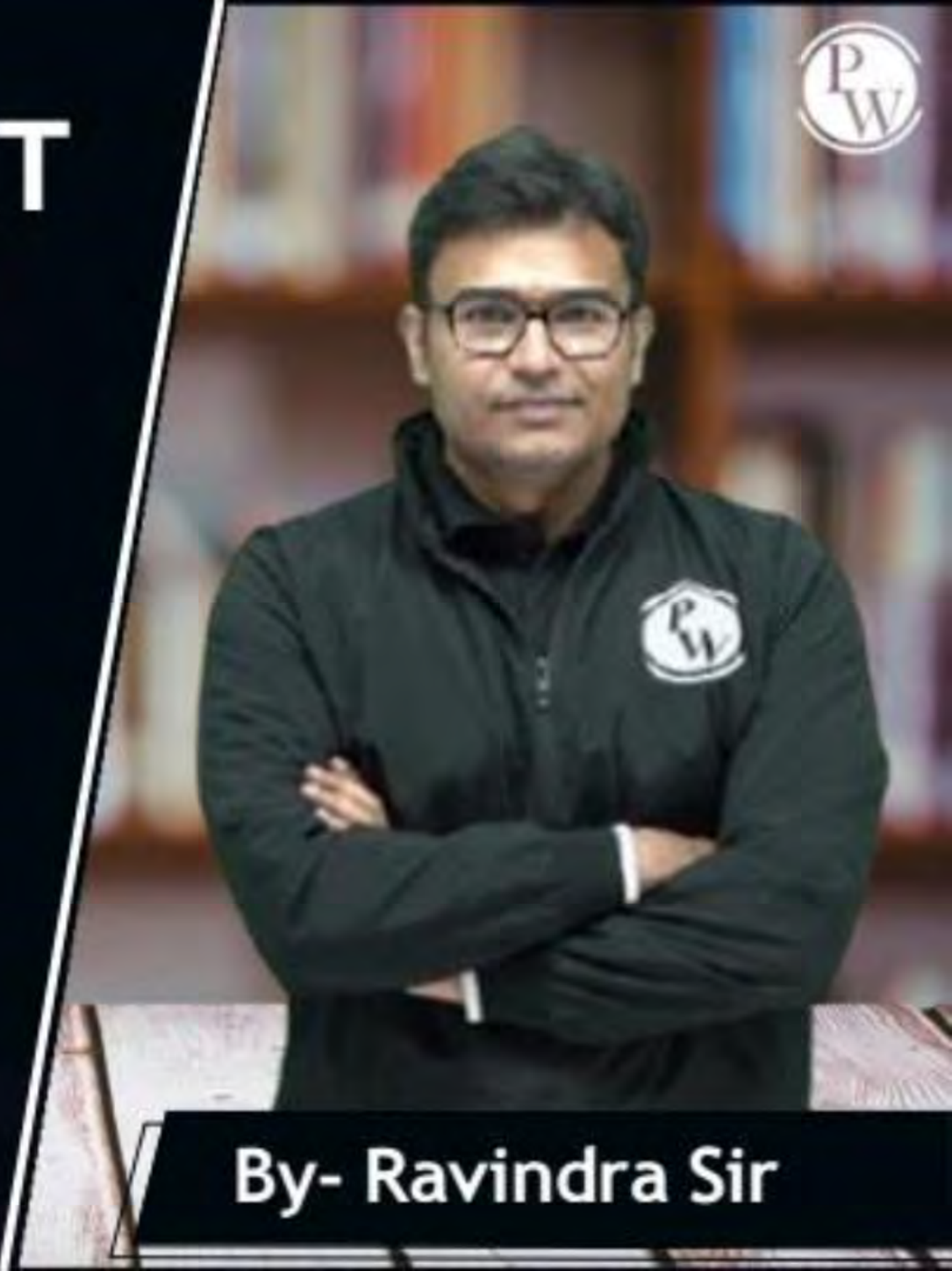
Computer Science & IT

COMPUTER NETWORK
(CN)

Routing

Lecture No. 1



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

ICMP

Topics to be Covered



- Topic
- Topic
- Topic

Routing

Inspiring Stories : Erik W.



Background: From USA. Blind since youth.

Struggle: Could not see mountains he loved.

Achievements: Climbed Everest and all Seven Summits.

Impact: Darkness never stopped him.

Inspiring Stories : Marlee Matlin



Background: **Deaf since a child.**

Struggle: **Rejected in acting world.**

Achievements: **Won an Oscar for acting.**

Impact: **Proved talent has no barriers.**

Inspiring Stories : Chen Zhou

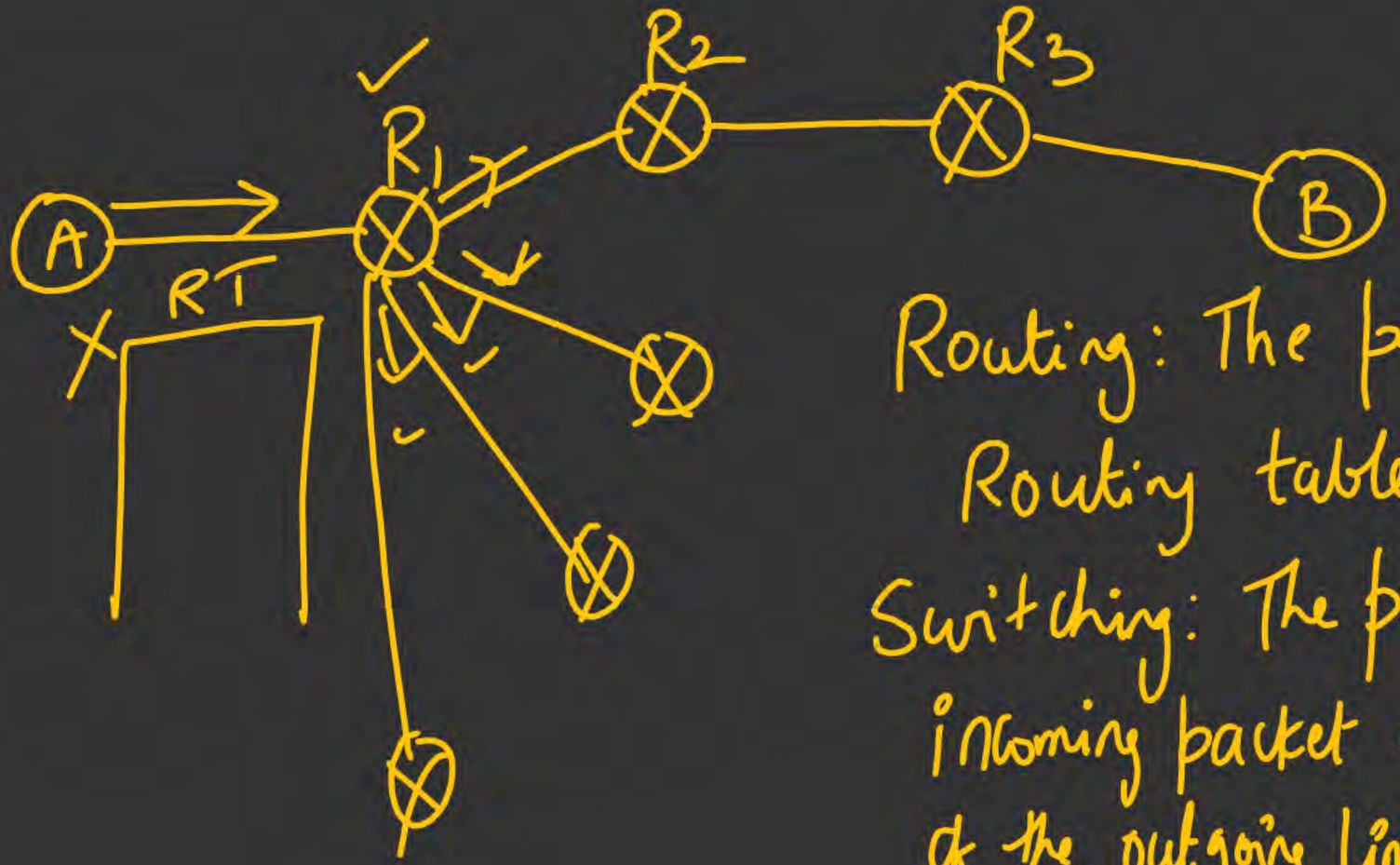


Background: From China. Lost both legs in train accident at 13.

Stuggle: Begged and sang on streets to live.

Achievements: Climbed China's five holy mountains with wooden supports.

Impact: Became a hero of willpower.



Routing: The process of preparing Routing table

Switching: The process of taking an incoming packet and placing it on one of the outgoing links

Flooding: Taking a packet from an incoming edge and placing it on all outgoing edges

Routing

Dis: (i) Routing tables are required

(ii) Processing time is required

(iii) Shortest path is not guaranteed

(iv) Reliability is less

Flooding

adv: (i) No routing

(ii) Shortest path is always guaranteed

(iii) Highly reliable

Routing

adv : (i) No duplicate packets
(ii) Traffic is less

flooding

dis : (i) Duplicate packets
(ii) Traffic is very high

If we want reliability and dont care about traffic, flooding is best

In military n/w, flooding is used

Routing algorithms

Static

→ If we know the entire graph of the routers and prepare Routing tables manually then it is Static algo

→ There is no central authority

→ Depending on traffic and topology, the tables will not change

Dynamic

→ Routers will prepare routing tables

→ Depending on traffic and topology the routing tables changes

→ No manual intervention is required

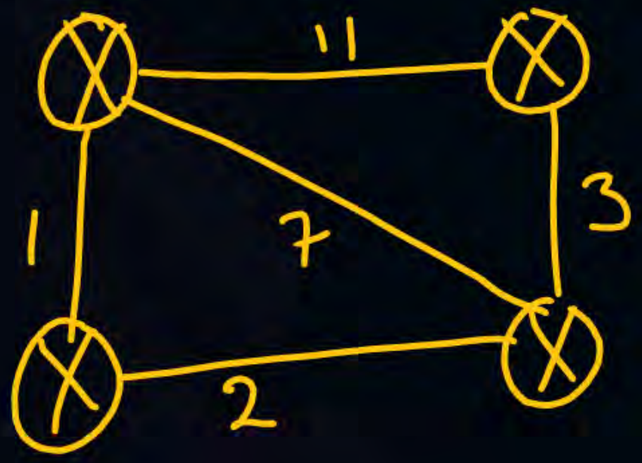
- DVR
- LSR

DVR (Distance vector routing)

↓

array

It is very popular in the initial days of the internet

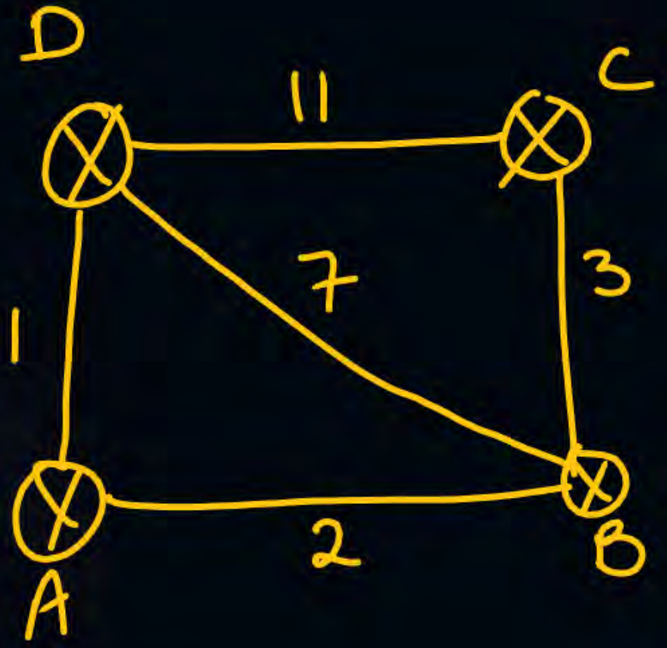


The numbers associated with edges can be cost, or distance or traffic or queuing delays

Step 1:- every router will construct local routing table using information about the neighbours

Destination	DIS	Next hop

Des	Dis	NH
A	1	A
B	7	B
C	11	C
D	0	D



Des	Dis	NH
A	∞	—
B	3	B
C	0	C
D	11	D

Des	Dis	NH
A	0	A
B	2	B
C	∞	—
D	1	D

DES	DIS	NH
A	2	A
B	0	B
C	3	C
D	7	D

Des	Dis	NH
A	1	A
B	7	B
C	11	C
D	0	D

Des	Dis	NH
A	∞	—
B	3	B
C	0	C
D	11	D



Des	Dis	NH
A	0	A
B	2	B
C	∞	—
D	1	D

Des	Dis	NH
A	2	A
B	0	B
C	3	C
D	7	D

Step 2:- every router will take its distance vector and sends it to its neighbour

A → D, B

B → A, C, D

C → D, B

D → A B C

Des	Dis	NH
A	1	A
B	7	B
C	11	C
D	0	D



Des	Dis	NH
A	∞	—
B	3	B
C	0	C
D	11	D

Des	Dis	NH
A	0	A
B	2	B
C	∞	—
D	11	D

Des	Dis	NH
A	2	A
B	0	B
C	3	C
D	7	D

At A :-

From B



From D



→ Single edge

⇝ path

Des	Dis	NH
A	0	A
B	2	B
C	5	B
D	1	D

$A \rightsquigarrow D = \min \{ A \xrightarrow{①} D + D \rightsquigarrow D,\ A \xrightarrow{②} B + B \rightsquigarrow D \}$

$A \rightsquigarrow B = \min \{ A \xrightarrow{①} D + D \xrightarrow{⑦} B,\ A \xrightarrow{②} B + B \xrightarrow{⓪} B \}$

$A \rightsquigarrow C = \min \{ A \xrightarrow{①} D + D \xrightarrow{11} C,\ A \xrightarrow{②} B + B \xrightarrow{③} C \}$

Des	Dis	NH
A	1	A
B	7	B
C	11	C
D	0	D



Des	Dis	NH
A	∞	—
B	3	B
C	0	C
D	11	D

Des	Dis	NH
A	0	A
B	2	B
C	∞	—
D	1	D

Des	Dis	NH
A	2	A
B	0	C
C	3	C
D	7	D

At A :-

From D

1
7
11
0

AD = 1

From B

2
0
3
7

AB = 2

Des	Dis	NH
A	0	A
B	2	B
C	5	B
D	1	D

Des	Dis	NH
A	1	A
B	7	B
C	11	C
D	0	D



Des	Dis	NH
A	∞	—
B	3	B
C	0	C
D	11	D

Des	Dis	NH
A	0	A
B	2	B
C	∞	—
D	11	D

Des	Dis	NH
A	2	A
B	0	B
C	3	C
D	7	D

A ± B

From A

0
2
∞
1

From C

∞
3
0
11

From D

1
7
11
0

BA = 2 BC = 3 BD = 7

Des	Dis	NH
A	2	A
B	0	B
C	3	C
D	3	A

Des	Dis	NH
A	1	A
B	7	B
C	11	C
D	0	D

Des	Dis	NH
A	∞	—
B	3	B
C	0	C
D	11	D



Des	Dis	NH
A	0	A
B	2	B
C	∞	—
D	11	D

Des	Dis	NH
A	2	A
B	0	B
C	3	C
D	7	D

At C

From D

1
7
11
0

CD = 11

From B

2
0
3
7

CB = 3

Des	Dis	NH
A	5	B
B	3	B
C	0	C
D	10	B

Des	Dis	NH
A	1	A
B	7	B
C	11	C
D	0	D

Des	Dis	NH
A	∞	—
B	3	B
C	0	C
D	11	D

Des	Dis	NH
A	0	A
B	2	B
C	∞	—
D	11	D

Des	Dis	NH
A	2	A
B	0	B
C	3	C
D	7	D



At D:-

From A	From B	From C
0	2	∞
2	0	3
∞	3	0
1	7	11

DA = 1 DB = 7 DC = 11

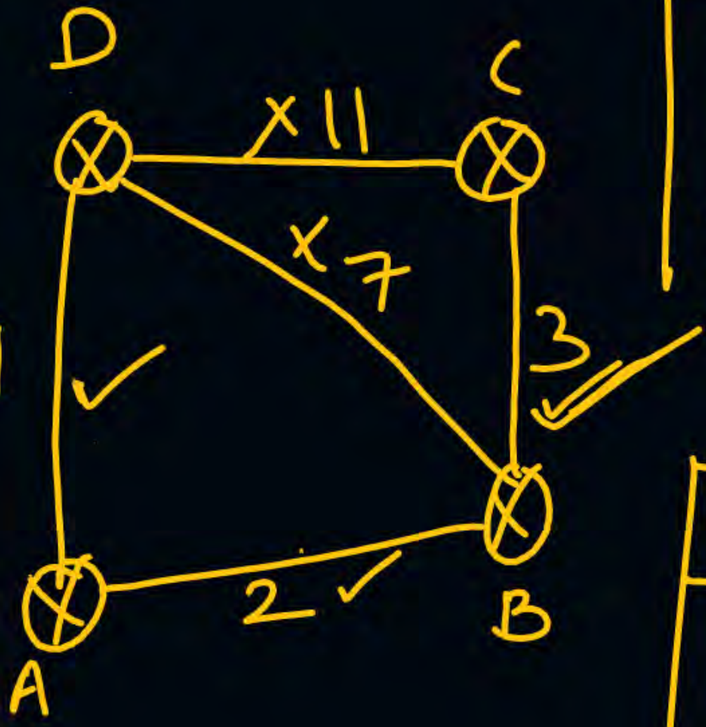
A	1	A
B	3	A
C	10	B
D	0	D

step 1: shortest path with contain max 1 edge

Step 2. " " " " 2 edges

DVR will keep on sending DVs periodically.

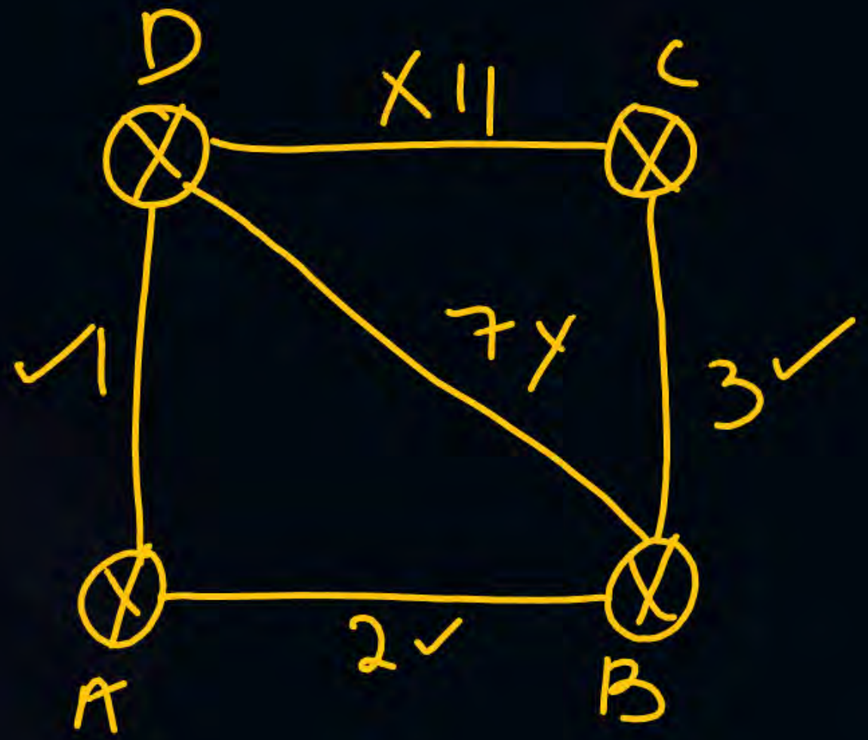
Des	Dis	NH
A	1	A
B	3	A
C	6	A
D	0	D



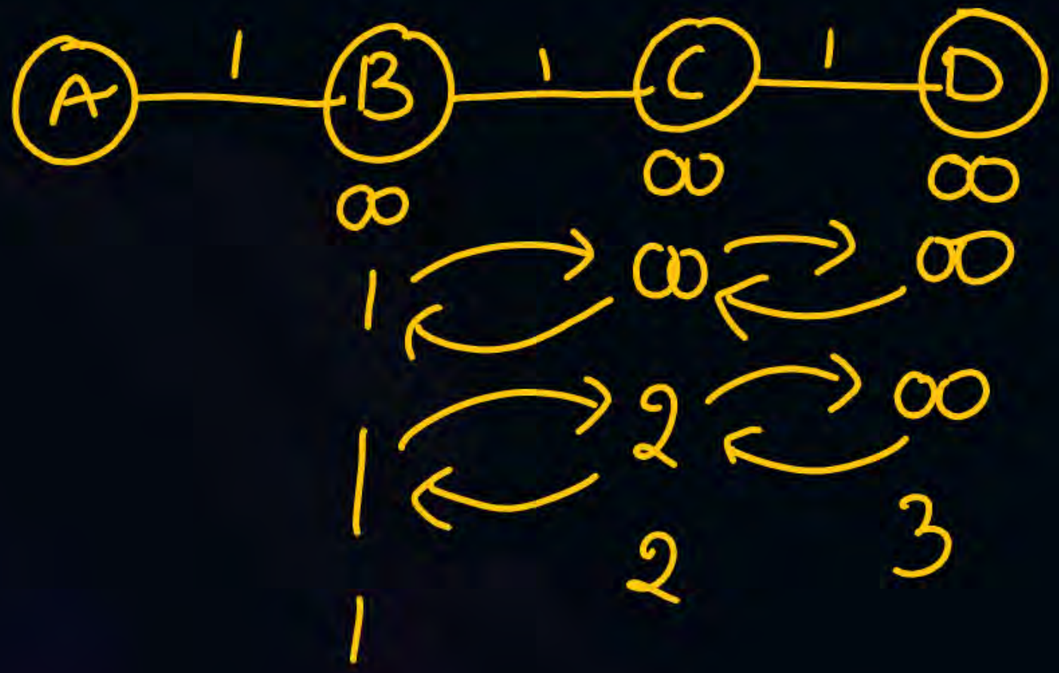
Dest	Dis	NH
A	5	B
B	3	C
C	0	C
D	6	B

Des	Dis	NH
A	0	A
B	2	B
C	5	B
D	1	A

Des	Dis	NH
A	2	BA
B	0	B
C	3	C
D	3	A


D
X 11
C
7x
1
3
2
A
B

Dis adv of DVR is count to ∞ (Bad new spreads slow
(good news spread fast





At N_3:-

From N_2

1 0 6 7 3

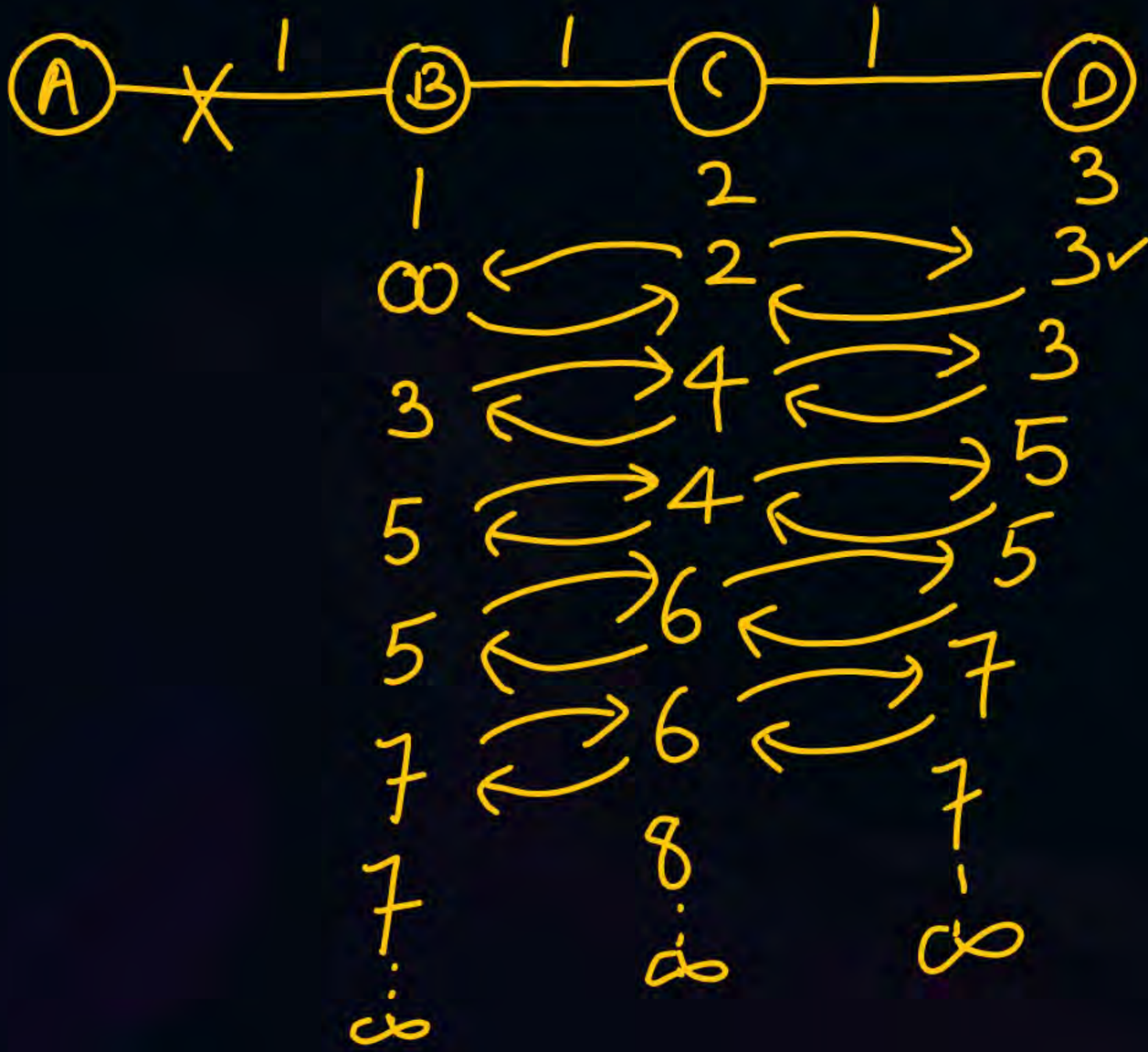
From N_4

8 7 2 0 4

$N_3N_2 = 2$ $N_3N_4 = 2$

N_1	3	N_2
N_2	2	N_2
N_3	0	N_3
N_4	2	N_4
N_5	5	N_2

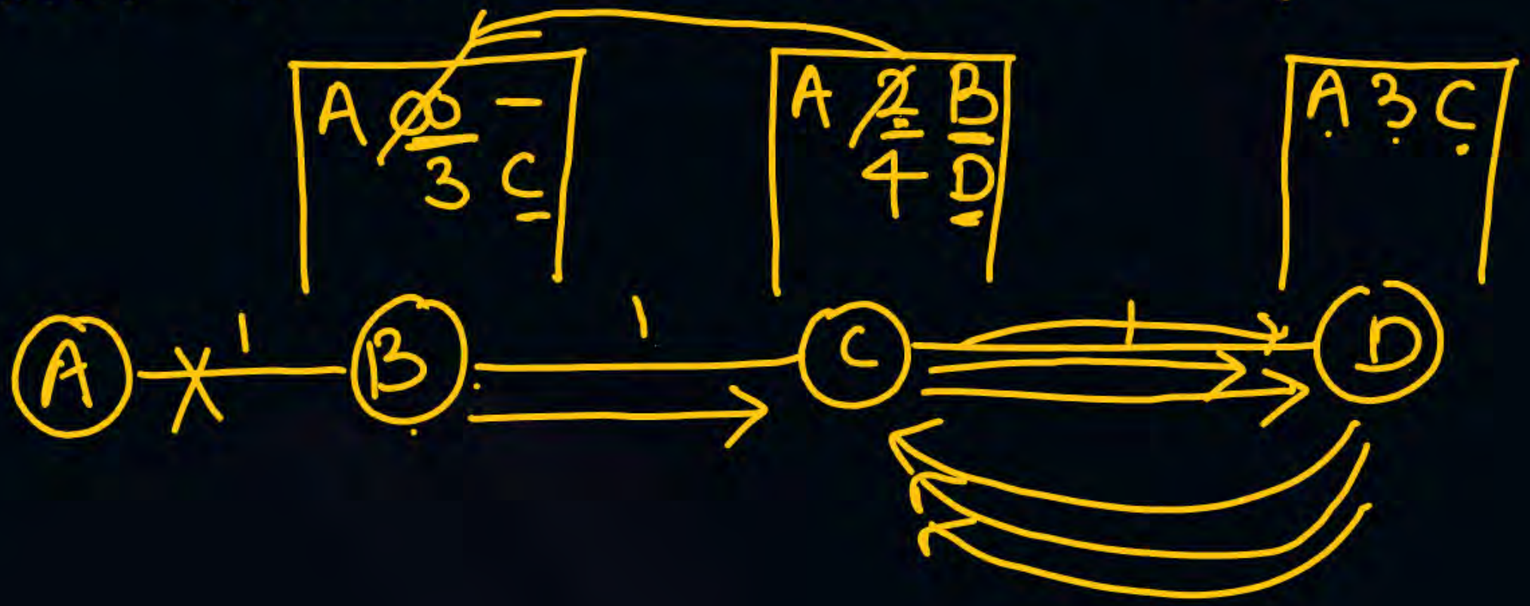
Disadv of DVR → Count to ∞



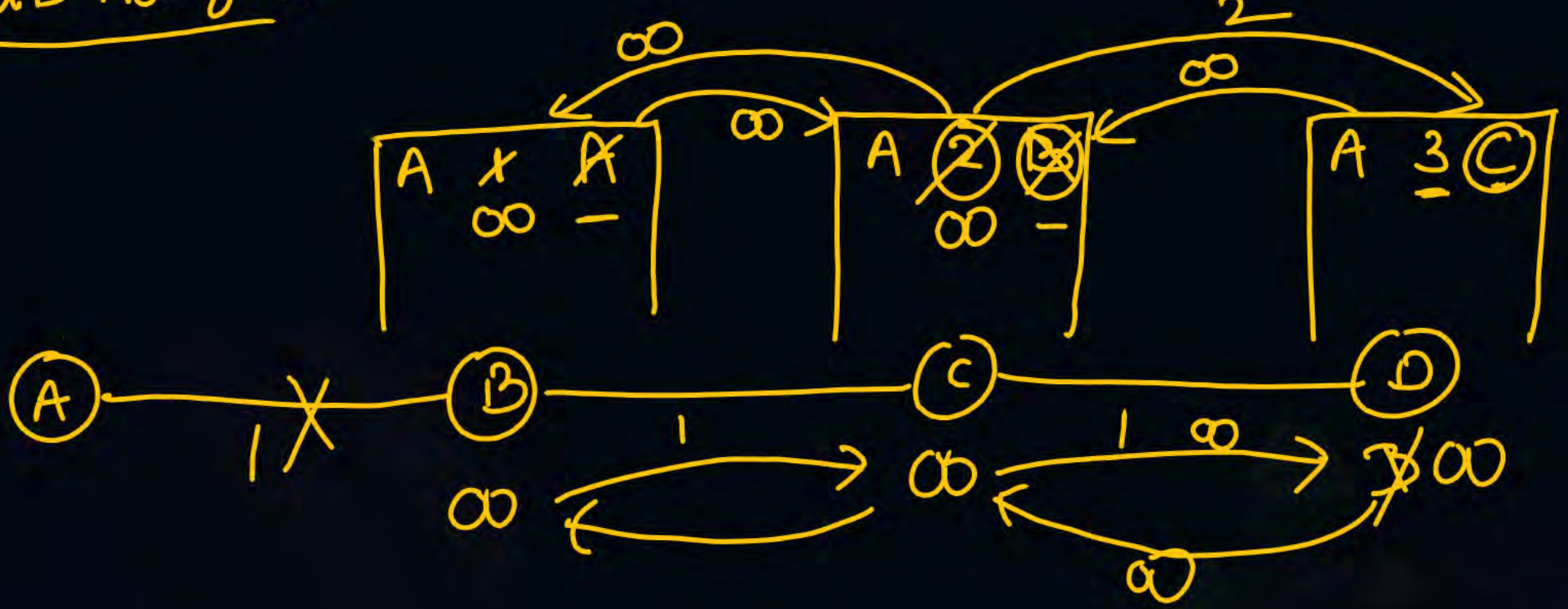
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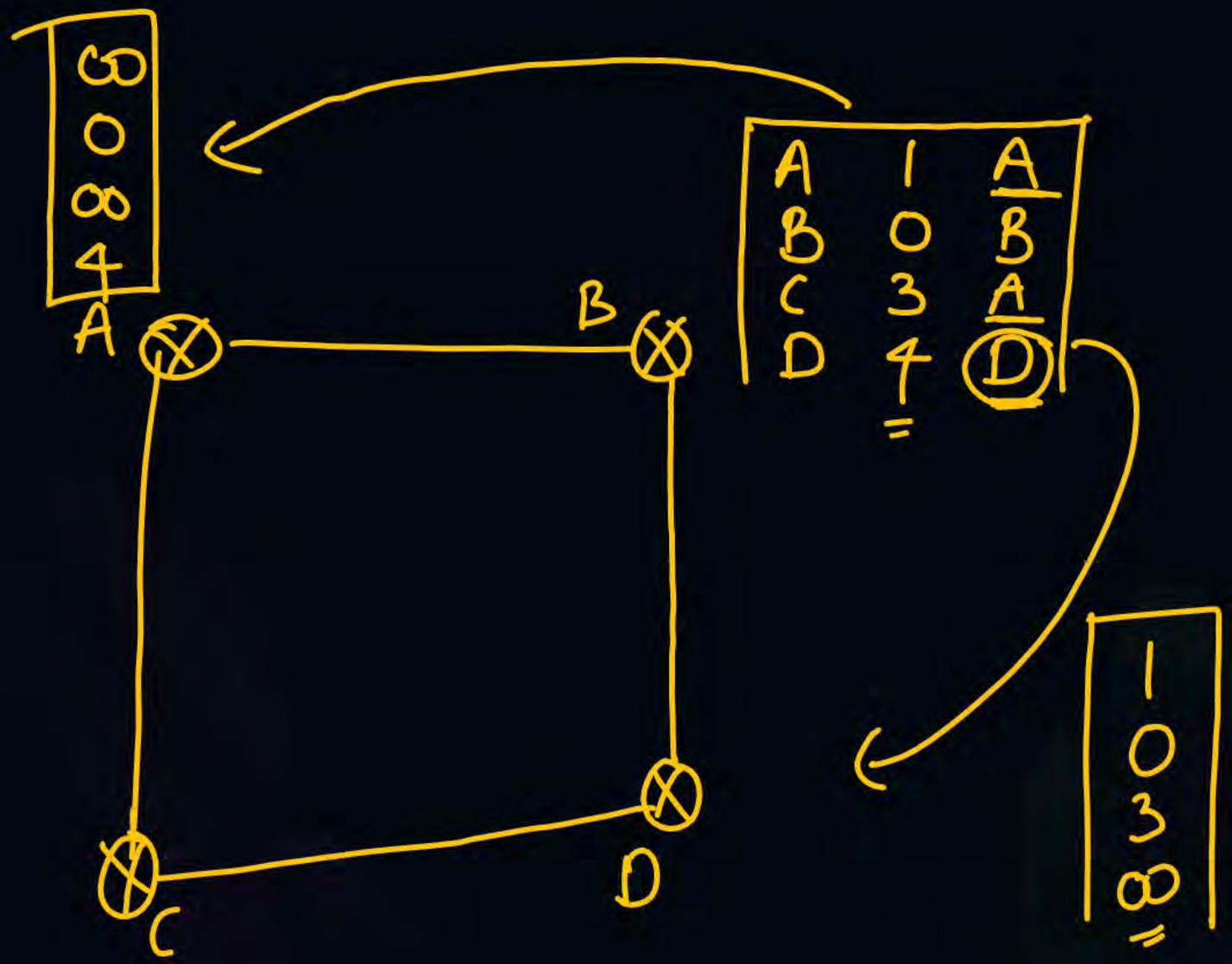
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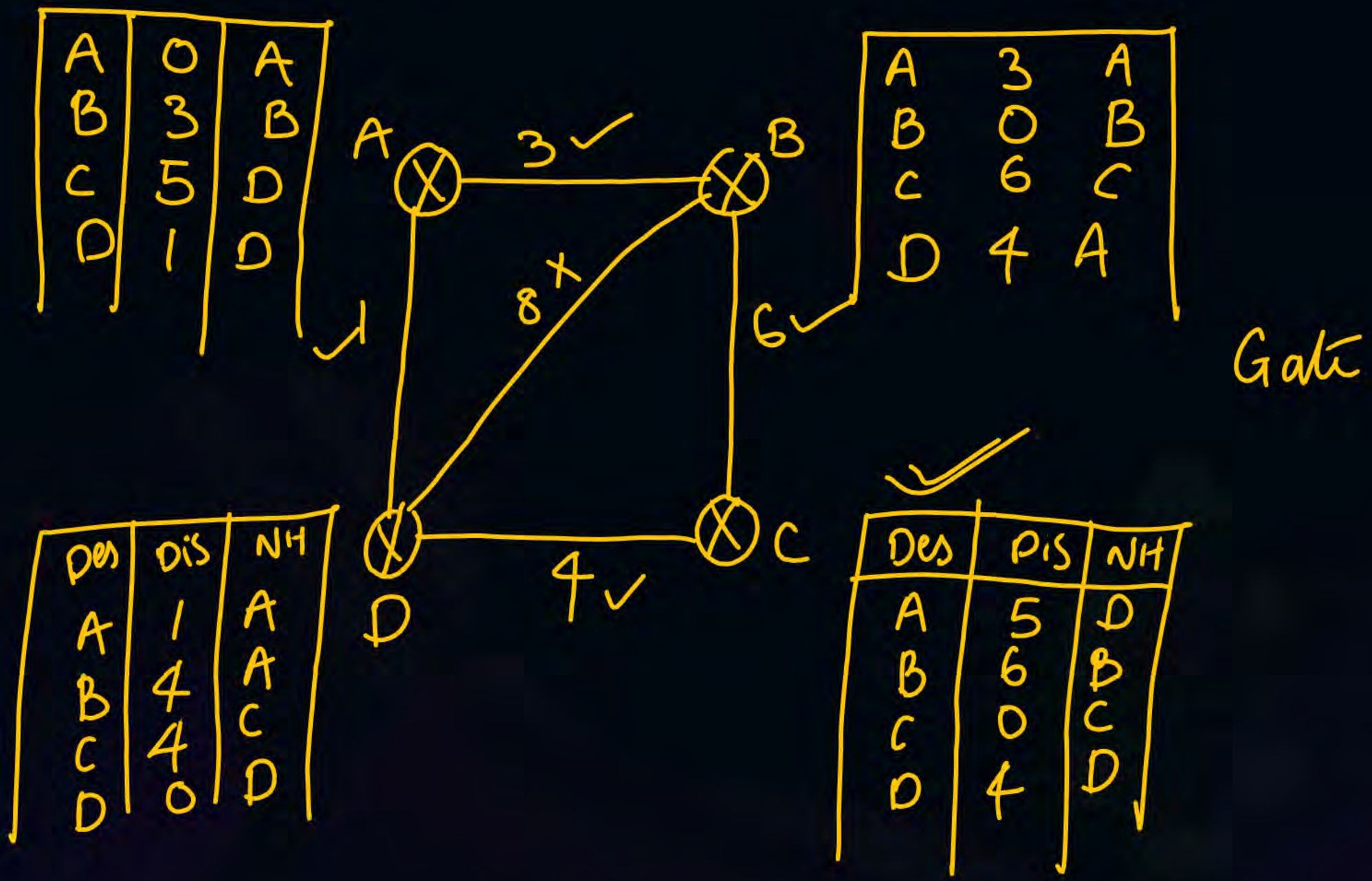
Count to ∞ Can introduce loops



Split horizon




0
0
∞
4
A
B
C
D
A 1 A
B 0 B
C 3 A
D 4 D
1
0
3
∞


A
B
C
D
3
8
6
4
1
Galt

THANK - YOU